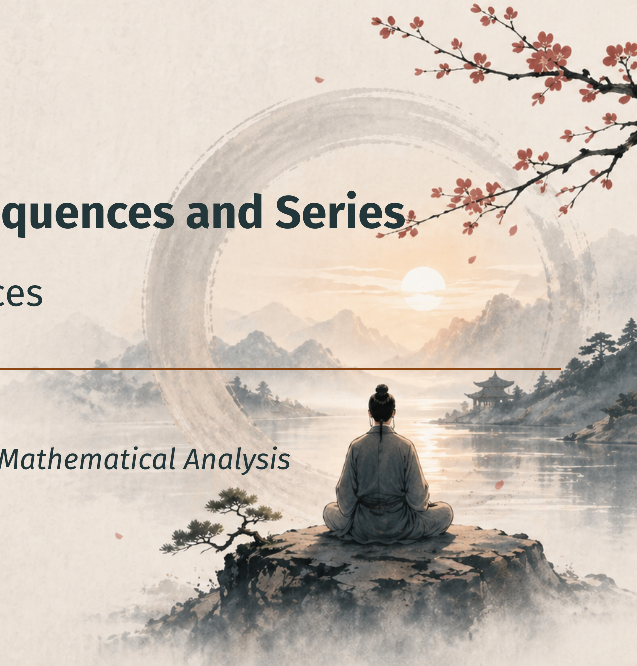
Chapter 3: Numerical Sequences and Series

Section: Convergent Sequences

Based on Walter Rudin's *Principles of Mathematical Analysis*



Outline

Motivation

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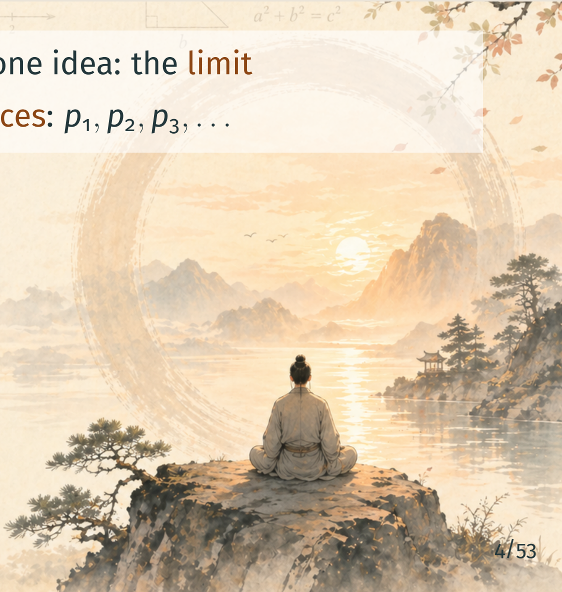


Why This Section Matters

- The whole of analysis is built on one idea: the limit
- The simplest limits live in sequences: $p_1, p_2, p_3, \ldots$
- Metric spaces give us the general framework: $d(p_n, p) \to 0$

Definition and Examples

Definition 3.1: Convergent Sequence

Definition 3.1 — Convergence

A sequence $\{p_n\}$ in a metric space X converges if there is a point $p \in X$ such that:

$$\forall \varepsilon > 0 \ \ \exists N \in \mathbb{N} \ \text{ with } \ n \geq N \implies d(p_n, p) < \varepsilon.$$

Definition 3.1: Notation and Terminology

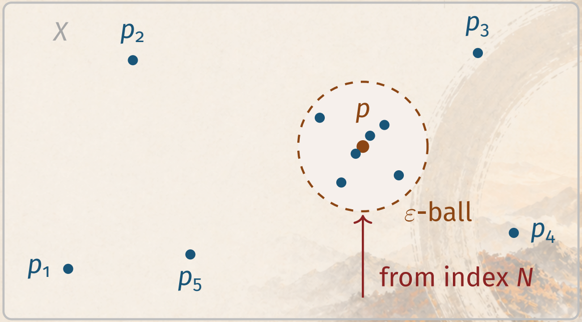
Definition 3.1 — Convergence

$\forall \varepsilon > 0 \;\; \exists N$ with $n \geq N \implies d(p_n, p) < \varepsilon$.

Notation and terminology:

- p is the limit of $\{p_n\}$; write $p_n \to p$ or $\lim_{n\to\infty} p_n = p$.
- If no such p exists, $\{p_n\}$ diverges.

Picture: What Convergence Looks Like



No matter how small the ε-ball around p, eventually every term lies inside.

Convergence Depends on the Space X

Observation

Convergence depends on both $\{p_n\}$ and the ambient metric space X.

Example: $s_n = 1/n$

- In $\mathbb{R}^1$: $s_n \to 0$ (converges)

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Convergence depends on both $\{p_n\}$ and the ambient metric space X.

Example: $s_n = 1/n$

- In $\mathbb{R}^1$: $s_n \to 0$ (converges)
- In $X = \{x \in \mathbb{R} : x > 0\}$: diverges (the candidate limit $0 \notin X$)

When in doubt, say "convergent *in* X."

Range and Boundedness

Range and Bounded Sequence

- The range of $\{p_n\}$ is the set $\{p_n : n = 1, 2, 3, \ldots\}$.
- $\{p_n\}$ is bounded if its range is a bounded set.

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- The range of $\{p_n\}$ is the set $\{p_n : n = 1, 2, 3, \ldots\}$.
- $\{p_n\}$ is bounded if its range is a bounded set.

Two independent features:

- *convergent or divergent*
- *bounded or unbounded*

Examples will show almost all combinations are possible, except: convergent $\Rightarrow$ bounded.

Examples 3.1: Five Sequences in $\mathbb{C}$

We work in $X = \mathbb{R}^2 = \mathbb{C}$.

Features to watch for each sequence:

- **Convergence** (yes or no, to what limit)
- **Range** (finite or infinite)
- **Boundedness** (bounded or unbounded)

Example 3.1 (a): $s_n = 1/n$

Sequence: $s_n = \dfrac{1}{n}$

$$1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$$

- Converges: $\lim_{n\to\infty} \frac{1}{n} = 0$
- Range: infinite
- Bounded: yes (all terms in $[0, 1]$)

Example 3.1 (b): $s_n = n^2$

Sequence: $s_n = n^2$

$$1,\ 4,\ 9,\ 16,\ 25,\ \ldots$$

- **Diverges:** grows without bound
- Range: infinite
- **Unbounded**

Example 3.1 (c): $s_n = 1 + (-1)^n/n$

Sequence: $s_n = 1 + \dfrac{(-1)^n}{n}$

$$0,\ \frac{3}{2},\ \frac{2}{3},\ \frac{5}{4},\ \frac{4}{5},\ \dots$$

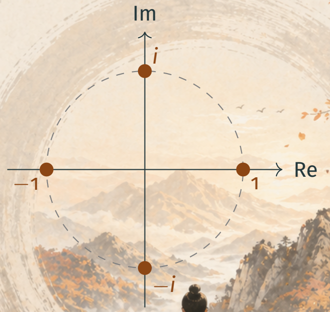
- Converges: $\lim_{n\to\infty} s_n = 1$
- Range: infinite
- Bounded

Example 3.1 (d): $s_n = i^n$

Sequence: $s_n = i^n$

$$i,\ -1,\ -i,\ 1,\ i,\ -1,\ \ldots$$

- Diverges: cycles forever
- Range: finite: $\{i, -1, -i, 1\}$
- Bounded: on unit circle



Example 3.1 (e): Constant Sequence $s_n = 1$

Sequence: $s_n = 1$ for all n

$$1,\ 1,\ 1,\ 1,\ 1,\ \ldots$$

- Converges: $\lim_{n\to\infty} s_n = 1$
- Range: finite: $\{1\}$
- Bounded

Lesson: The range of a convergent sequence may still be finite.

Properties of Convergent Sequences

Theorem 3.2: Four Properties of Convergent Sequences

Let $\{p_n\}$ be a sequence in a metric space X.

Theorem 3.2

(a) $p_n \to p \iff$ every $N_\varepsilon(p)$ contains p_n for all but finitely many n.

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(c) $\{p_n\}$ convergent $\implies$ $\{p_n\}$ bounded.

(d) $E \subset X$, p a limit point of $E \implies \exists \{p_n\} \subset E$ with $p_n \to p$.

Proof of 3.2(a): Forward Direction

(a): $p_n \to p \iff$ every $N_\varepsilon(p)$ contains p_n for all but finitely many n.

$\Rightarrow$ direction

Suppose $p_n \to p$ and let $V = N_\varepsilon(p)$ be a neighborhood of p, $\varepsilon > 0$.
Then $d(q,p) < \varepsilon \Rightarrow q \in V$.

By definition, $\exists\, N$ with

$$n \geq N \implies d(p_n, p) < \varepsilon \implies p_n \in V.$$

So V contains p_n for all $n \geq N$: all but finitely many.

Proof of 3.2(a): Converse

(a): $p_n \to p \iff$ every $N_\varepsilon(p)$ contains p_n for all but finitely many n.

$\Leftarrow$ direction

Suppose every $N_\varepsilon(p)$ contains all but finitely many of the p_n.

Fix $\varepsilon > 0$. Set $V = N_\varepsilon(p) = \{q \in X : d(p,q) < \varepsilon\}$ (the ε-ball).

By assumption, $\exists\, N$ with $p_n \in V$ for $n \geq N$.

Hence $d(p_n, p) < \varepsilon$ for $n \geq N$, so $p_n \to p$. $\blacksquare$

Proof of 3.2(b): Uniqueness — Setup

(b): If $p_n \to p$ and $p_n \to p'$, then $p = p'$.

Setup

Suppose $p_n \to p$ and $p_n \to p'$. Fix $\varepsilon > 0$.

Pick N, N' so that

$$n \geq N \implies d(p_n, p) < \tfrac{\varepsilon}{2},$$
$$n \geq N' \implies d(p_n, p') < \tfrac{\varepsilon}{2}.$$

Strategy: use the triangle inequality to bound $d(p, p')$ by ε.

Proof of 3.2(b): Uniqueness — Triangle Inequality

(b): If $p_n \to p$ and $p_n \to p'$, then $p = p'$.

Triangle inequality

Let $n \geq \max(N, N')$. Then

$$d(p, p') \;\leq\; d(p, p_n) + d(p_n, p') \;<\; \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon.$$

Conclusion

$d(p, p') < \varepsilon$ for every $\varepsilon > 0 \implies d(p, p') = 0 \implies \boxed{p = p'.}$ ■

Proof of 3.2(c): Convergent $\Rightarrow$ Bounded — Setup

(c): $\{p_n\}$ convergent $\implies$ $\{p_n\}$ bounded.

Setup

Suppose $p_n \to p$. Take $\varepsilon = 1$: $\exists\, N$ with

$$n > N \implies d(p_n, p) < 1.$$

So the *tail* $p_{N+1}, p_{N+2}, \ldots$ fits in the unit ball around p.

What about the initial segment $p_1, \ldots, p_N$? Only finitely many points.

Proof of 3.2(c): Convergent $\Rightarrow$ Bounded — Conclusion

(c): $\{p_n\}$ convergent $\implies$ $\{p_n\}$ bounded.

Bounded radius

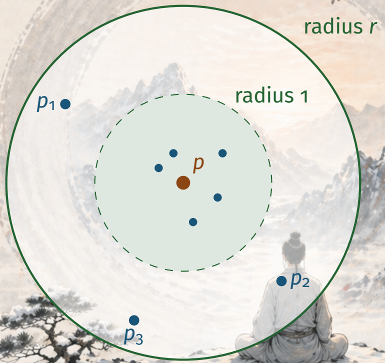
Define

$$r = \max\{\, 1,\ d(p_1, p),\ \ldots,\ d(p_N, p)\,\}.$$

Then $d(p_n, p) \le r$ for every n.

Conclusion

$\{p_n\}$ sits inside the closed ball of radius r around p. ■



Proof of 3.2(d): Limit Point $\Rightarrow$ Sequential Limit

(d): If p is a limit point of $E \subset X$, then $\exists \{p_n\} \subset E$ with $p_n \to p$.

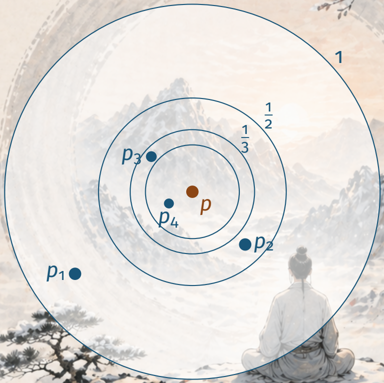
Construction

p is a limit point of E: every $N_\varepsilon(p)$ meets $E \setminus \{p\}$.

For each $n = 1, 2, 3, \ldots$ the ball $N_{1/n}(p)$ meets E.

Pick $p_n \in E$ with

$$d(p_n, p) < \frac{1}{n}.$$



Proof of 3.2(d): Verifying Convergence

(d): If p is a limit point of $E \subset X$, then $\exists \{p_n\} \subset E$ with $p_n \to p$.

Verify $p_n \to p$

Given $\varepsilon > 0$, by the Archimedean property choose N with $N\varepsilon > 1$.

For $n > N$:

$$d(p_n, p) < \frac{1}{n} < \frac{1}{N} < \varepsilon.$$

Hence $p_n \to p$. $\blacksquare$

Consequence: topology meets sequences — limit points *are* sequential limits from within E.

Arithmetic of Limits

From Metric Spaces to Complex Sequences

Context

Theorem 3.2 used only the metric d. Now we add algebraic structure:

Complex sequences $\{s_n\}, \{t_n\} \subset \mathbb{C}$ can be *added, multiplied, divided*.

Question: Do limits respect these operations?

Yes — and the proofs are classical ε-style estimates.

Theorem 3.3: Arithmetic of Limits

Suppose $s_n \to s$ and $t_n \to t$ in $\mathbb{C}$. Then:

Theorem 3.3

(a) $\lim\limits_{n\to\infty} (s_n + t_n) = s + t$

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(b) For any $c \in \mathbb{C}$: $\lim\limits_{n\to\infty} cs_n = cs$, $\lim\limits_{n\to\infty} (c + s_n) = c + s$

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(c) $\lim\limits_{n\to\infty} s_n t_n = st$

(d) $\lim\limits_{n\to\infty} \dfrac{1}{s_n} = \dfrac{1}{s}$, provided $s_n \neq 0$ and $s \neq 0$.

Proof of 3.3(a): Sum

The split-epsilon trick

Given $\varepsilon > 0$, pick N_1, N_2 so that

$$n \geq N_1 \Rightarrow |s_n - s| < \tfrac{\varepsilon}{2}, \quad n \geq N_2 \Rightarrow |t_n - t| < \tfrac{\varepsilon}{2}.$$

For $n \geq N = \max(N_1, N_2)$:

$$|(s_n + t_n) - (s + t)| \leq |s_n - s| + |t_n - t| < \tfrac{\varepsilon}{2} + \tfrac{\varepsilon}{2} = \varepsilon.$$

$$\boxed{s_n + t_n \;\to\; s + t.} \quad \blacksquare$$

Note on 3.3(b): Scalar and Constant

Constant addition: $\lim(c + s_n) = c + s$

The constant sequence $t_n := c$ converges to c (every term equals the limit). Apply 3.3(a) to s_n and t_n.

Scalar multiplication: $\lim c\,s_n = c\,s$

If $c = 0$, both sides equal 0. Otherwise, given $\varepsilon > 0$, pick N with

$$n \geq N \implies |s_n - s| < \frac{\varepsilon}{|c|}.$$

Then $|c\,s_n - c\,s| = |c|\,|s_n - s| < \varepsilon.$ ■

Proof of 3.3(c): Product — The Key Identity

Algebraic identity

$$s_n t_n - st = (s_n - s)(t_n - t) + s(t_n - t) + t(s_n - s). \tag{1}$$

Idea: each of the three pieces on the right can be driven to 0.

- $(s_n - s)(t_n - t)$ is quadratically small.
- $s(t_n - t)$ and $t(s_n - s)$ are linearly small.

Proof of 3.3(c): Controlling the Quadratic Piece

First piece $\to 0$

Given $\varepsilon > 0$, pick N_1, N_2 so that

$$n \geq N_1 \Rightarrow |s_n - s| < \sqrt{\varepsilon}, \qquad n \geq N_2 \Rightarrow |t_n - t| < \sqrt{\varepsilon}.$$

For $n \geq \max(N_1, N_2)$:

$$|(s_n - s)(t_n - t)| < \sqrt{\varepsilon} \cdot \sqrt{\varepsilon} = \varepsilon.$$

Therefore $\lim_{n\to\infty} (s_n - s)(t_n - t) = 0$.

Proof of 3.3(c): Assembling the Product Rule

Assemble (1)

Looking at identity (1): $s_n t_n - st$ is a sum of three terms.

- $(s_n - s)(t_n - t) \to 0$ (just shown)
- $s(t_n - t) \to s \cdot 0 = 0$ (by 3.3(b))
- $t(s_n - s) \to t \cdot 0 = 0$ (by 3.3(b))

By 3.3(a), the sum of three null sequences is null:

$$\lim_{n\to\infty} (s_n t_n - st) = 0 \implies \boxed{s_n t_n \to st.} \quad \blacksquare$$

Proof of 3.3(d): Reciprocal — Bounding Away from 0

Step 1 — Keep $|s_n|$ away from 0

Since $s \neq 0$, apply convergence with $\frac{1}{2}|s|$:

$$\exists\, m \text{ with } n \geq m \Rightarrow |s_n - s| < \tfrac{1}{2}|s|.$$

Reverse triangle inequality $\Rightarrow |s_n| > \frac{1}{2}|s|$:

$$|s_n| = |s + s_n - s| \geq |s| - |s_n - s| > \tfrac{1}{2}|s| \quad (n \geq m).$$

Why this matters: $1/s_n$ only makes sense if $|s_n|$ stays away from 0.

Proof of 3.3(d): Reciprocal — The Estimate

Step 2 — Estimate the reciprocal

Given $\varepsilon > 0$, pick $N > m$ so that

$$n \geq N \Rightarrow |s_n - s| < \tfrac{1}{2}|s|^2\varepsilon.$$

Then for $n \geq N$:

$$\left|\frac{1}{s_n} - \frac{1}{s}\right| = \left|\frac{s_n - s}{s_n s}\right| < \frac{2}{|s|^2}|s_n - s| < \varepsilon.$$

So $\boxed{\dfrac{1}{s_n} \to \dfrac{1}{s}.}$ ■

Convergence in $\mathbb{R}^k$

Moving Up to $\mathbb{R}^k$

From $\mathbb{C}$ to $\mathbb{R}^k$

A sequence in $\mathbb{R}^k$ is a sequence of vectors $\mathbf{x}_n = (\alpha_{1,n}, \alpha_{2,n}, \dots, \alpha_{k,n})$.

Natural question: What is the relationship between

- convergence of the *vector* sequence $\{\mathbf{x}_n\}$, and
- convergence of each *coordinate* sequence $\{\alpha_{j,n}\}$?

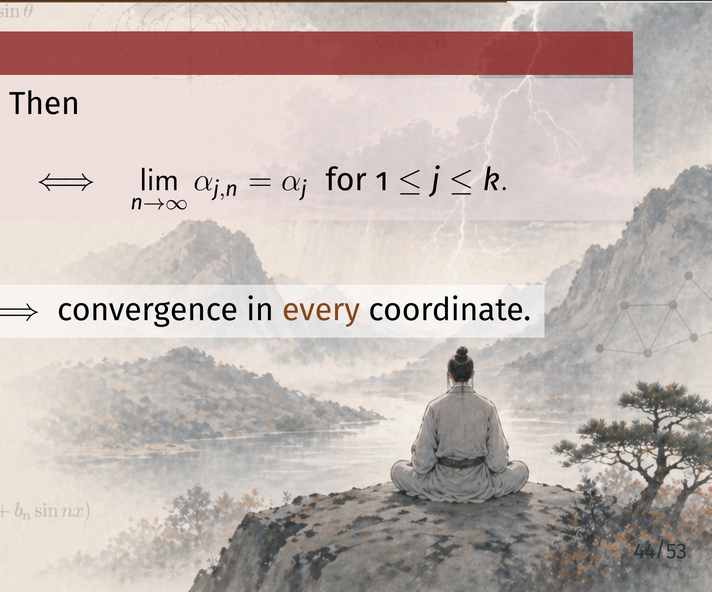
Theorem 3.4: Componentwise Convergence

Theorem 3.4 (a)

Let $\mathbf{x}_n = (\alpha_{1,n}, \ldots, \alpha_{k,n}) \in \mathbb{R}^k$. Then

$$\mathbf{x}_n \to \mathbf{x} = (\alpha_1, \ldots, \alpha_k) \iff \lim_{n\to\infty} \alpha_{j,n} = \alpha_j \text{ for } 1 \le j \le k.$$

Vector convergence $\iff$ convergence in every coordinate.



Theorem 3.4: Algebraic Operations in $\mathbb{R}^k$

Theorem 3.4 (b)

If $\mathbf{x}_n \to \mathbf{x}$, $\mathbf{y}_n \to \mathbf{y}$ in $\mathbb{R}^k$ and $\beta_n \to \beta$ in $\mathbb{R}$, then

$$\mathbf{x}_n + \mathbf{y}_n \to \mathbf{x} + \mathbf{y},$$

$$\mathbf{x}_n \cdot \mathbf{y}_n \to \mathbf{x} \cdot \mathbf{y},$$

$$\beta_n \mathbf{x}_n \to \beta \mathbf{x}.$$

Limits commute with sum, dot product, and scalar multiplication.

Proof of 3.4(a): $\Rightarrow$ Direction

From vector convergence to coordinate convergence

Suppose $\mathbf{x}_n \to \mathbf{x}$ in $\mathbb{R}^k$. For each fixed j, a single squared term is at most the full sum of squares:

$$|\alpha_{j,n} - \alpha_j|^2 \;\le\; \sum_{i=1}^{k} |\alpha_{i,n} - \alpha_i|^2 \;=\; |\mathbf{x}_n - \mathbf{x}|^2.$$

Taking square roots:

$$|\alpha_{j,n} - \alpha_j| \;\le\; |\mathbf{x}_n - \mathbf{x}|.$$

So $|\mathbf{x}_n - \mathbf{x}| \to 0 \implies |\alpha_{j,n} - \alpha_j| \to 0$ for every j.

Proof of 3.4(a): $\Leftarrow$ Direction

From coordinate convergence to vector convergence

Suppose $\alpha_{j,n} \to \alpha_j$ for each $1 \le j \le k$. Given $\varepsilon > 0$, pick N so that $n \ge N$ implies

$$|\alpha_{j,n} - \alpha_j| < \frac{\varepsilon}{\sqrt{k}} \qquad (1 \le j \le k).$$

Then

$$|\mathbf{x}_n - \mathbf{x}| = \left\{ \sum_{j=1}^{k} |\alpha_{j,n} - \alpha_j|^2 \right\}^{1/2} < \sqrt{k \cdot \frac{\varepsilon^2}{k}} = \varepsilon.$$

Hence $\mathbf{x}_n \to \mathbf{x}$. ■

Proof of 3.4(b): Operations Preserve Limits

By reduction to 3.3

Write $\mathbf{x}_n \cdot \mathbf{y}_n = \sum_{j=1}^{k} \alpha_{j,n}\beta_{j,n}$ where $\mathbf{y}_n = (\beta_{1,n}, \ldots, \beta_{k,n})$.

By 3.4(a), $\alpha_{j,n} \to \alpha_j$ and $\beta_{j,n} \to \beta_j$.

By 3.3(a)+(c), each $\alpha_{j,n}\beta_{j,n} \to \alpha_j\beta_j$, and the sum of k such terms tends to $\mathbf{x} \cdot \mathbf{y}$.

Sum and scalar multiplication follow similarly. $\blacksquare$

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Key takeaways from this section:

- Definition 3.1: $p_n \to p$ if $\forall\,\varepsilon > 0, \exists\, N$ with $n \geq N \Rightarrow d(p_n, p) < \varepsilon$.

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- Theorem 3.2: neighborhood form, *uniqueness*, bounded, limit-point bridge.
- Theorem 3.3: limits of sums, products, and reciprocals for $\mathbb{C}$-sequences.
- Theorem 3.4: convergence in $\mathbb{R}^k$ is *componentwise*, and operations commute with limits.

Coming up next: *Subsequences* — when subsequences converge, and the powerful notion of *subsequential limits*.